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PAPER

Transmission Raman measurement directly through packed corn kernels to improve sample representation and accuracy of compositional analysis

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The potential of transmission Raman spectroscopy for direct analysis of packed granular samples, one of the most frequently encountered sample types in the field of non-destructive spectroscopic analysis, has been evaluated. For this purpose, transmission Raman spectra were collected by laser illumination through packed corn kernels to determine their protein concentration. Back-scattering Raman spectra of the same samples were also collected for comparison. Raman spectral features of the major kernel constituents were initially characterized, and Raman mapping over the whole kernel face was performed to investigate the inhomogeneous distribution of constituents in a kernel. Possible variations of transmission spectral features depending on the laser illumination on different locations of a kernel were investigated, since the orientation of kernels in the packing was essentially random. Rotation of kernel packing during spectral collection was helpful in improving the compositional representation of packed kernels. With partial least squares (PLS) regression, the protein concentrations were determined using both spectral collection methods and the resulting accuracies were compared. As a result, the transmission measurement provided a more accurate determination of protein concentration since it enabled deeper sampling across the packed kernels, leading to a better compositional representation of them. By contrast, in the back-scattering measurement, kernels on the top of the packing were mainly sampled for the spectral acquisition. Moreover, the back-scattering spectral feature, more weighted to constituents localized at the outer portion of a kernel, was short of representing the overall composition of a kernel.

Introduction

Transmission Raman spectroscopy has recently emerged as a potential method for non-destructive analysis of diverse solid samples such as pharmaceutical tablets since it could alleviate the sub-sampling problem[1–6] that often occurs with back-scattering Raman measurements. In particular, for samples that are expected to be internally heterogeneous or composed of multiple layers, transmission Raman measurement is the obvious method of choice to ensure or at least improve representation of the entire sample. Recent developments and applications of transmission Raman spectroscopy have been reviewed in several publications.[7–15]

Beyond simple transmission Raman measurement of a granular sample individually, direct transmission analysis of packed granular samples is of further benefit, since a large amount of granules are often necessary to measure simultaneously to improve sample representation for a batch of samples due to possible compositional variability among individual granules. Practically such packed granular samples are frequently encountered for non-destructive spectroscopic analysis especially in the field of agriculture and polymer.[16,17] With back-scattering measurements, collected Raman spectra are mainly representative of granular samples only on the top of packing, rather than the whole. By contrast, transmission measurement is able to obtain Raman spectra pertinent to granular samples across the packing, so a larger number of granules could be simultaneously sampled at a given spectral collection. Therefore, it could be beneficial to acquire spectra with a better representation for a batch of samples.

More importantly, when granular samples are internally heterogeneous in the distribution of constituents, the back-scattering spectral feature would be more weighted with chemical information of constituents localized at near the surface. By contrast, transmission measurement enables to produce Raman spectra representing the overall composition of constituents in a granule better regardless of their localization. Despite these merits of transmission Raman measurement for direct analysis of packed granular samples, it has not been readily explored so far.

We evaluated the analytical potential of transmission Raman measurement for the analysis of packed granular samples, corn kernels in this study, and attempted to determine their protein concentration. Protein concentration of corns as well as many

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other agricultural products is mostly determined by the Kjeldahl method, requiring an extensive sample preparation for acid digestion and a long analysis time for titration. On the contrary, vibrational spectroscopic methods such as transmission Raman measurement are non-destructive and fast when properly implemented.

Corn kernels are internally composed of several constituents[18] including endosperm, germ, pericarp (bran coat) and tip cap as shown in Fig. 1(a). Initially, Raman spectra of each major constituent were collected and the resulting spectral features were compared. To investigate the inhomogeneous distribution of the constituents inside a kernel, Raman mapping covering the whole kernel was performed. Furthermore, the orientation of each kernel in the packing with respect to the direction of laser illumination is not consistent. Therefore, possible variations of the transmission spectral features of kernels, when laser is illuminated at different locations on the kernels, were investigated.

For quantitative analysis, transmission Raman spectra of kernels were collected directly through the packing. For comparison, back-scattering Raman spectra were also collected for the same samples. Since it was expected that the sample representation by both measurements for packed kernels would differ, the corresponding spectral features were closely examined. Further, the reproducibility of replicate spectra collected with random kernel packing was evaluated for both measurements. Finally, with the collected back-scattering and transmission spectra, partial least squares (PLS) regression was used to determine the protein concentration, and the resulting accuracies were evaluated.

Experimental

Corn samples and Raman spectral collection

Forty-five corn samples of multiple international origins were obtained from a Korean feed company. Reference protein concentrations were determined by the Kjeldahl method at the company that provided the samples. For mapping, a kernel was sectioned along the wide face as shown in Fig. 1(a). The sectioned kernel was positioned on a microscope stage connected with a Raman spectrometer (Kaiser Optical Inc., $\lambda_{ex} = 785$ nm) and a laser beam (approximately 20 μm in diameter) was focused on the face using an objective lens (10×/0.25 numerical aperture) for spectral collection (exposure time: 10 s, 10 scans).

Transmission Raman spectra were collected by direct illumination of laser radiation (785 nm, Invictus, Kaiser Optical Inc.) onto the packed kernels housed in a 15 cm diameter glass Petri dish (exposure time: 10 s, 60 scans). The thickness of the packing was approximately 1.5 cm. The Petri dish packed with kernels was rotated during spectral collection at 15 rpm. The diameter of the laser illumination spot was approximately 1.5 mm, and the power of the laser beam at the packing was 200 mW. Transmitted Raman signals were collected at the opposite direction from laser illumination using a wide area illumination (WAI) scheme (PhAT system, Kaiser Optical Inc.).[19] Back-scattering Raman spectra were also collected for the same corn samples (exposure time: 10 s, 10 scans). The diameter of the laser illumination was 6 mm, and the back-scattered Raman signal was collected using the same instrumentation (WAI scheme). Spectra for both measurements were collected in triplicate for each sample at room temperature. For each spectral collection, kernels were reloaded into the Petri dish to ensure random packing. The resolution of both transmission and back-scattering spectra was 4 cm^{-1}. All calculations and multivariate analyses, including baseline correction, intensity normalization, principal component analysis (PCA) and PLS,

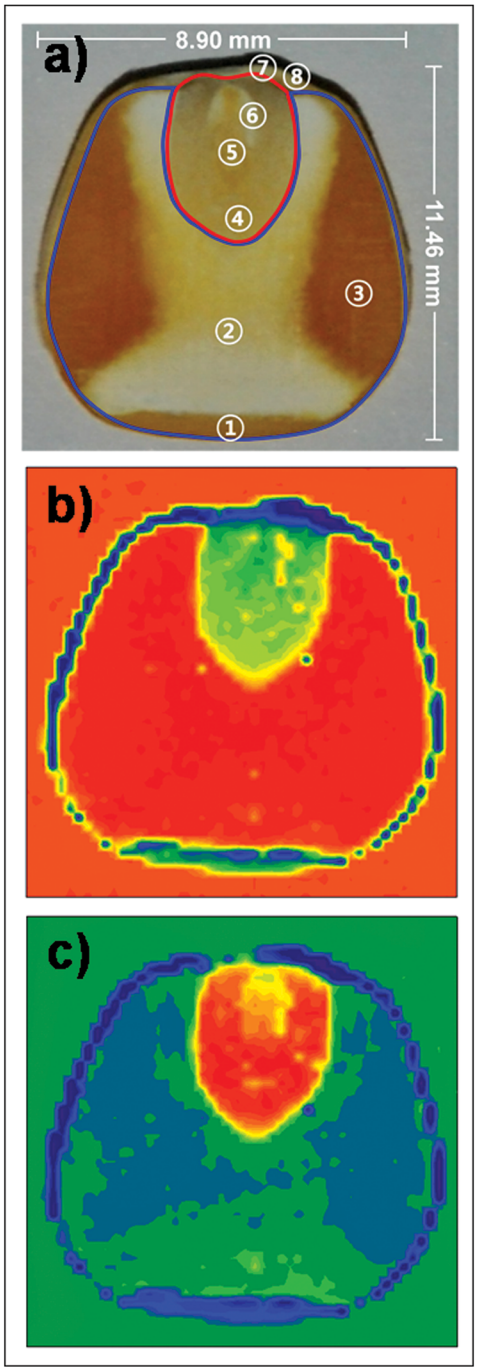

Fig. 1 Picture of a kernel sectioned along its wide face is shown (a). The numbered circles on the picture indicate 8 different locations where Raman spectra were collected. The sections designated by the blue and red lines correspond to endosperm and germ, respectively. Raman mapping images of the kernel constructed using the second (b) and third (c) score are shown.

were conducted using MATLAB version 7.0 software (The Math-Works Inc., MA, USA).

Results and discussion

Raman spectral features of a corn kernel

As is generally known, a corn kernel is composed of several internal constituents that include the endosperm, germ, pericarp and tip cap.[18] Using a microscope, Raman spectra were collected at 8 different locations of a sectioned kernel as indicated by the numbered circles on the picture in Fig. 1(a). Spectra (1780–390 cm^{-1} range) acquired at the circles are shown in Fig. 2. The spectra were normalized and arbitrarily offset for clear comparison. For normalization, the baselines of raw spectra were linearly corrected at 1780, 1505, 1220, 970, 822, and 390 cm^{-1}, and the area under the 1780–390 cm^{-1} range was calculated. Then, each baseline-corrected spectrum was divided by the corresponding area for normalization. For comparison, Raman spectra of corn starch, zein, and hull are also shown in the figure. Endosperm, corresponding to the section of a kernel designated by the blue line in Fig. 1(a), occupies approximately 80 wt% (hereinafter %) of a kernel and consists predominantly of starch (~88%).[18] Thus, the spectral features acquired at ①, ② and ③ were similar and closely resembled the spectrum of corn starch (95% purity, purchased in a local market). The section highlighted by the red line corresponds to the germ (~11% of a kernel) and is composed of diverse components such as oil (~30%), protein (~18%), fiber (~18%), and others (starch and ash).[18] Therefore, the spectral features acquired at ④, ⑤ and ⑥ fairly vary compared to those collected at the endosperm. For example, spectrum ⑤ has a relatively stronger intensity at 1003 cm^{-1}. The spectrum of zein, a typical protein in corn, possesses a band at 1003 cm^{-1}, thus spectrum ⑤ would be acquired where the protein was more localized. Similarly, spectrum ⑥ shows a stronger starch band at 480 cm^{-1}, again indicating spectral acquisition where starch was more localized. The spectral features acquired at ⑦ and ⑧ are nearly analogous to those acquired from the hull, prepared by carefully scratching off only the outer surfaces of kernels. Overall, Raman spectral features obtained from the internal constituents were largely different as expected.

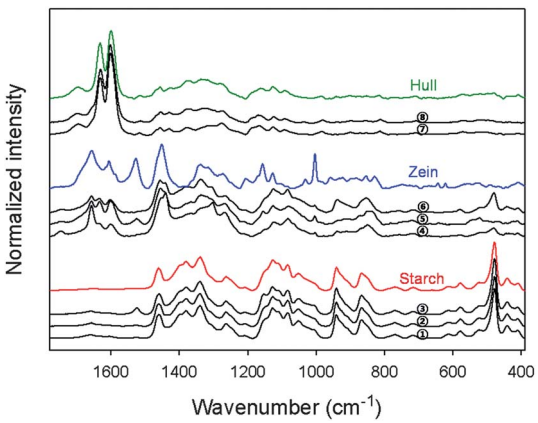

Fig. 2 Raman spectra collected at 8 different locations on a kernel as indicated by the numbered circles on the picture in Fig. 1(a). Raman spectra corresponding to three pure constituents of corn (starch, zein and hull) are shown. The spectra are arbitrarily offset for comparison.

The protein content of corn is normally from 5 to 9%.[18] Since protein is widely present throughout a kernel (3–6% in endosperm, 2.0–2.5% in germ, 0.2% in pericarp), the collection of Raman spectra covering the entire kernel is necessary to accurately account for the total protein content. Raman spectra acquired by laser illumination on a localized section of the kernel could be insufficient to be fully representative of the chemical composition of the entire kernel and inaccurate for quantitative compositional analysis.

Identification of inhomogeneous distribution of constituents inside a kernel by Raman mapping

To further characterize the inhomogeneous distribution of constituents inside a kernel on a microscopic scale, Raman mapping covering a whole kernel was performed (mapping area: 1.4 cm^2 (1.2 cm × 1.2 cm)). The sectioned kernel shown in Fig. 1(a) was used again for mapping. The point-to-point distance in both the x and y directions was 250 μm, resulting in a total of 2500 spectra (50 × 50 points) for the mapping. PCA was performed using all of the spectra in the mapping and the scores were used to build images. Spectra were normalized as described previously before performing PCA. Fig. 1(b) and (c) show Raman images of the kernel constructed using the second and third scores, respectively.

In the second score image, endosperm (red), germ (green and yellow) and pericarp (blue) were clearly distinguishable. In the third score image, differentiation between floury endosperm (green) and vitreous endosperm (bluish green) corresponding to locations ② and ③ in Fig. 1(a), respectively, is fairly distinct. Although the spectral features of these two constituents were quite similar as shown in Fig. 2, minute spectral differences, such as around 1524 cm^{-1}, led to discrimination of these two constituents in the third score image. In addition, compositional variation inside the germ was noticeable, as indicated by the yellow/orange color. Overall, images constructed with the scores show distinct compositional heterogeneity in the kernel. The spectral features described by each factor (principal component) were different as confirmed by the two score images, implying that the adoption of several factors would be necessary to detail the compositional variability in the kernel.

Transmission Raman spectral features of kernels

The constituents of corn are distinctly localized as shown in Fig. 1 and the shape of each kernel was close to that of a flat disk. Therefore, depending on the position of laser illumination on a kernel for spectral collection, the resulting transmission Raman features would possibly vary. This is based on the fact that, to produce transmission Raman spectra, the actual kernel-covered volume relative to the whole kernel volume was inconsistent when the orientation of the kernel varied with the fixed laser illumination direction.

To investigate the possible variation of the transmission spectral feature, the laser was targeted to illuminate 12 different spots on a kernel as shown in Fig. 3, and the acquired transmission features in the 1780–390 cm^{-1} range were examined. The

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numbered red circles and red arrows in the pictures indicate the laser illuminated spots and the illumination direction, respectively. In this investigation, the laser was illuminated on 4 different faces of the kernel (referred to as case I, II, III and IV). Raman spectra collected at ①, ② and ③ (hereinafter designated as spectra ①, ② and ③, respectively) were acquired by laser illumination along the vertical center-line on the wide face of the kernel (case I). For collection of spectra ④, ⑤ and ⑥, laser was focused on the opposite wide face along the vertical center-line (case II). Spectra ⑦, ⑧ and ⑨ were collected by laser illumination from the left to the right side of the kernel (case III). Similarly, spectra ⑩, ⑪ and ⑫ were acquired by laser irradiation from the top to the bottom of the kernel (case IV). At each illumination spot, triplicate spectra were collected. The size of laser illumination was 1.5 mm in diameter. Transmitted Raman signals crossing the kernel were collected also using the WAI scheme. This evaluation was repeated three times using three separate kernels, so 9 spectra (3 spectra per kernel) were obtained at each spot. For the purpose of comparison, back-scattering spectra were also collected at the same spots on the kernels. Careful attention was made to irradiate laser at the same position on each kernel; however, it would not be practically perfect. Moreover, the internal distribution of constituents in individual kernels is always inconsistent. Therefore, some degree of the kernel-to-kernel spectral variation would be expected among spectra collected at the same location on three different kernels.

Fig. 4(a) shows transmission spectra collected at 12 different spots on three individual kernels as described in Fig. 3. The spectra were arbitrarily offset for clear comparison. Each spectrum in the figure corresponds to an average of triplicate spectra collected at each spot. In case I, spectra ①, ② and ③ are moderately different from each other, while spectra ③ (red color) are more distinctly different from the others. Analogously the features of spectra ④, ⑤ and ⑥ in case II are also different from each other, while those of spectra ⑥ (red color) are substantially dissimilar to those of spectra ④ and ⑥. This is because both transmission spectra ③ and ⑥ were collected by directly illuminating through the germ, which had different compositions compared to those of the endosperm. For example, the intensity of the starch band at 480 cm^{-1} is considerably lower for spectra ③ and ⑥ compared to those of other spectra. For cases I and II, the sample volume covered for each spectral collection was relatively small compared to the entire volume of the kernel, although isotropically diffused laser radiation inside the kernel could increase the sample coverage volume to some degree. As a result, the spectra collected in both cases could contain localized compositional information of the kernel, thus providing an explanation for the different features of spectra ③ and ⑥ compared to the others.

In case III, the laser travel distance crossing the kernel became longer and laser radiation could diffuse more widely inside the kernel. Therefore, the overall sample coverage volume could be increased and so relatively less spectral variation would be expected regardless of the illuminated locations. As expected, the degree of spectral variation among the spectra in case III is relatively smaller compared to those in cases I and II. However, the features of spectra ⑨ (red color) are considerably different from those of spectra ⑦ and ⑧ especially in the 1700–1500 cm^{-1} range, since the laser illumination was directed more to the location where the germ was localized as shown in Fig. 3. The laser travel distance in case IV was slightly longer than that in case III, so the spectra collected in case IV would be most reproducible in terms of band shape and intensity due to the increased sample coverage volume. As a result, the magnitude of spectral variation among the spectra in case IV is least and the corresponding spectral features are relatively similar to each other.

Fig. 4(b) shows back-scattering spectra collected at 12 different spots on the same kernels. The most significant observation is that the band intensities of all back-scattering spectra in the 1700–1500 cm^{-1} range are substantially higher compared to those in the transmission spectra. These bands are characteristic peaks of hull, so clearly indicative of the more weighted sampling on the outer part of a kernel in the case of back-scattering

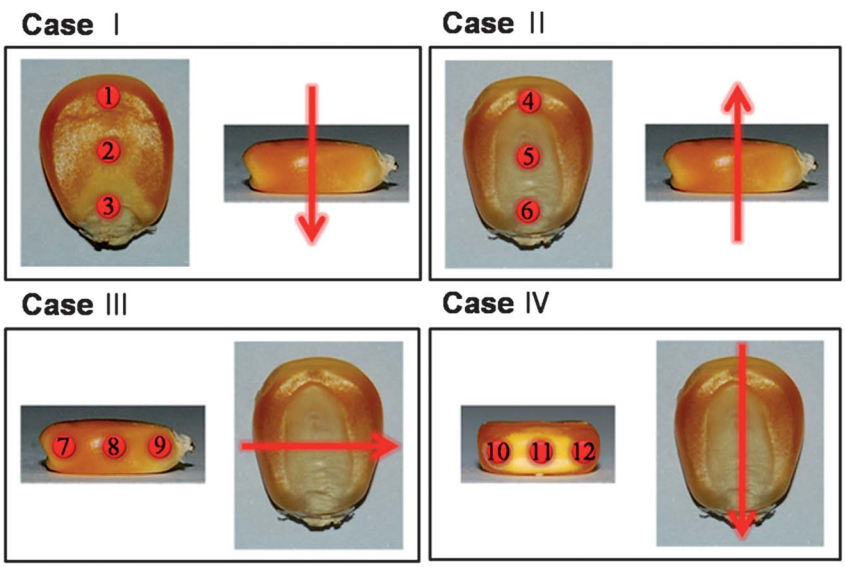

Fig. 3 Pictures indicating 12 different illuminated spots on a kernel for spectral collection. The numbered red circles and red arrows in the pictures indicate the laser illuminated spots and the illumination direction, respectively.

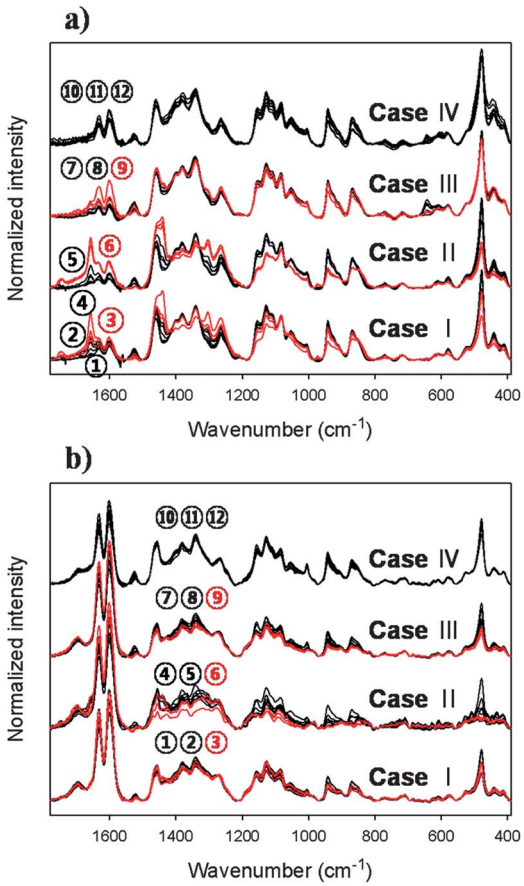

Fig. 4 Transmission (a) and back-scattering (b) spectra collected at 12 different locations described in Fig. 3. All spectra collected from 3 different kernels are shown. The spectra collected at each case are arbitrarily offset for comparison.

measurement. While the corresponding spectral features in case II are dissimilar to those of spectra in other cases, since the illuminated spots (④, ⑤ and ⑥) target clearly different kernel constituents. So, the resulting back-scattering spectral features such as the starch band at 480 cm^{-1} could sensitively vary depending on the illuminated spots in this case. By contrast, the spectral features collected in other cases are quite similar to each other, although minor spectral variation among the spectra in each case is observed. This result again demonstrates the more weighted sampling on the outer part of a kernel when back-scattering measurement is employed.

To simultaneously visualize all transmission and back-scattering spectra shown in Fig. 4, PCA was performed using these spectra. For easier comparison, the scores corresponding to transmission and back-scattering spectra are separately presented in Fig. 5(a) and (b), respectively. Both score scatter plots are constructed using the second and third scores, and the numbers indicate the corresponding spectra shown in Fig. 4. As shown in Fig. 5(b), the scores of spectra ④, ⑤ and ⑥ (case II) are fairly apart from the other scores in the case of back-scattering measurement. The scores from cases I, III and IV fairly aggregate each other and the aggregation indicates the similar spectral features as presented in Fig. 4(b). In the case of transmission measurement, the score variation is wider as expected (refer to Fig. 5(a)). The scores of spectra ③ and ⑥ are located relatively on the right, since the corresponding spectral features are different due to the more representation of the germ. Similarly, the scores (⑩, ⑪ and ⑫) from case IV are relatively positioned on the bottom left because the corresponding spectra are relatively more representative of the whole kernel, so the resulting spectral features could be somewhat different from those of other spectra.

The transmission spectra ① and ④ were collected by illuminating laser at nearly opposite spots on both faces and both illuminations aim endosperm mainly, so the corresponding scores are closely located to each other. The scores of spectra ② and ⑤ are quite nearly positioned to each other compared to the overall score distribution, while these are relatively more scattered compared to those of spectra ① and ④. This trend is more dominant for the scores of spectra ③ and ⑥.

The advantage of transmission measurement in this study is its deep sampling capability of across a kernel, while its sample representation would not be uniform across a kernel. As observed in spectra ③ and ⑥, transmission spectra could contain information relatively more representative of constituents localized around the front face, such as the germ in the case of ⑥ in Fig. 3, although the information of the endosperm located behind the germ is still obtainable by transmission measurement. This possibly explains the difference in transmission spectral features between spectra ③ and ⑥.

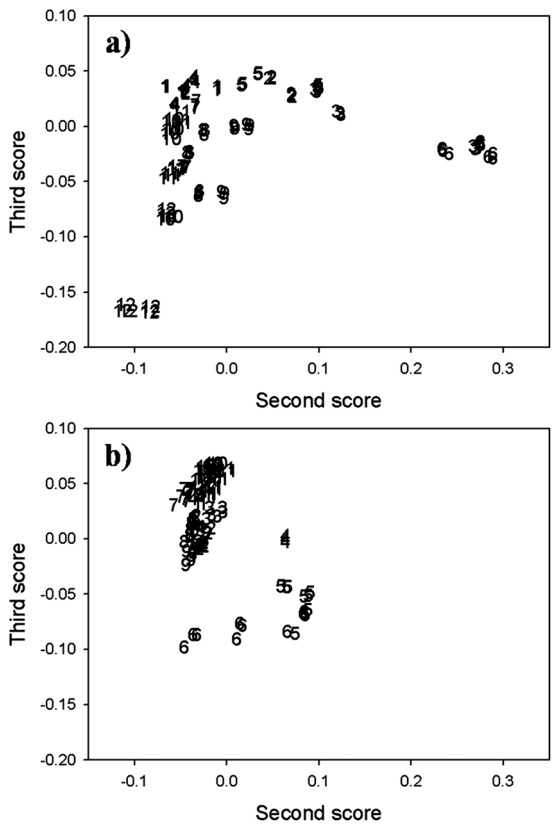

Fig. 5 The score scatter plots corresponding to transmission (a) and back-scattering (b) spectra in Fig. 4. Both score scatter plots are constructed using the second and third scores. The numbers indicate the corresponding spectra in Fig. 4.

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Further, when a kernel face is tilted with respect to the direction of laser illumination, the resulting transmission spectral feature would also vary further as found in a previous publication, in which spectral features of a tablet were examined by artificially tilting it.[20] Constituents in a kernel are heterogeneously distributed, so its transmission spectral feature could also sensitively vary when it is tilted.

In reality, packed granular samples have been directly measured for a better compositional representation of a batch, as routinely performed in non-destructive spectroscopic analysis.[17] For packed kernels, the orientation of each kernel in the packing with respect to the direction of laser illumination is not consistent. Therefore, overall transmission Raman spectral features of packed kernels would vary depending on the individual orientations of kernels in the packing. To minimize the possible spectral variation depending on the kernel packing, transmission Raman spectra are necessary to collect with continuous rotation of the packing to cover a larger amount of kernels. By this way, the resulting spectral feature could be more reproducible and representative of the overall composition of packed kernels.

Transmission Raman spectral features of packed kernels

As described above, Fig. 6(a) shows back-scattering and transmission Raman spectra of packed kernels collected with packing rotation at 15 rpm. The thickness of the packing was approximately 1.5 cm. Two to four kernels were packed across the packing depending on the individual orientations. Six replicate spectra were collected by reloading of the kernels at each spectral collection to ensure random packing. The overall spectral features acquired from both spectral collection methods were similar, while the baselines of the spectra varied vertically due to differences in the kernel packing for each spectral collection.

For a detailed examination, both normalized spectra were compared as shown in Fig. 6(b), with the black and red colors designating back-scattering and transmission spectra, respectively. The most noticeable difference between the spectra is the higher intensity of the bands at 1633 and 1600 cm^{-1} in the back-scattering spectra. As confirmed in Fig. 2, these bands were most dominant in the spectrum of the hull. This finding again indicates that the back-scattering spectra contain more weighted information on constituents localized at the outer part of the kernel, such as the hull. The intensity of the same bands in transmission spectra is relatively lower, indicating the decreased weight on constituents at the outer part of the kernels.

The reproducibility of six replicate spectra was investigated by observing the spectral features highlighted in the 1480–1420 and 1170–1140 cm^{-1} ranges, as shown in Fig. 6(c) and (d), respectively. In the 1480–1420 cm^{-1} range, the shapes of the transmission bands are quite reproducible, while those of the back-scattering spectra vary, such as the 1460 cm^{-1} band. The bands in the 1170–1140 cm^{-1} range are also more reproducible in the case of transmission measurement. For accurate compositional analysis, replicate spectra collected with random kernel packing should be reproducible as well as representative. Since the transmission spectral collection was accomplished through a 1.5 cm thickness packing, Raman spectra representative of multiple kernels located in the transmission beam path could be obtained. Due to the capability of deeper sampling across the kernel packing, the collected transmission Raman spectra could be more representative of the overall composition of kernels in the packing as well as more reproducible even with random kernel packing.

On the other hand, in back-scattering measurements, kernels located at the top of the packing were mainly sampled for spectral collection, and moreover, the faces of individual kernels exposed to laser illumination vary significantly during spectral collection. Consequently, the back-scattering spectral features of

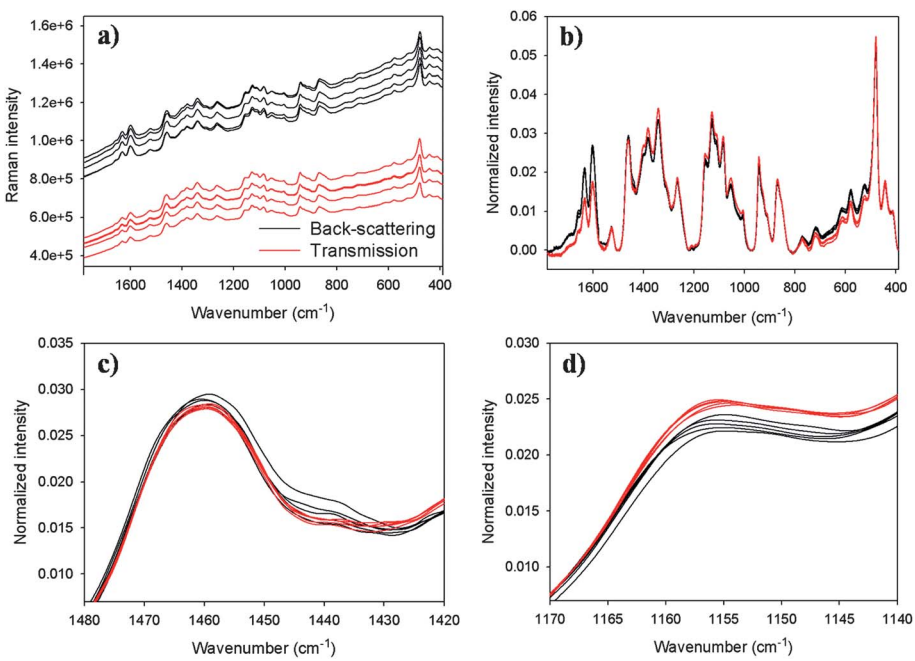

Fig. 6 Uncorrected back-scattering (black) and transmission (red) Raman spectra of packed kernels (a) and the corresponding normalized spectra (b). For a detailed examination, the spectral features in the 1480–1420 cm^{-1} (c) and 1170–1140 cm^{-1} (d) ranges are highlighted.

packed kernels would be more sensitive to the random packing as confirmed in Fig. 6 and insufficient to accurately represent the composition of kernels in the packing.

Determination of protein concentrations

With both back-scattering and transmission spectra acquired from the same corn samples, partial least squares (PLS) regression was used to determine the protein concentrations. The 1780–390 cm^{-1} range was used for PLS. Out of 45 total samples, 36 and 9 samples were randomly assigned to calibration and validation sets, respectively. The optimum number of factors was identified as the number of factors that gave a minimum standard error of cross-validation (SECV). Cross-validation was performed by dividing the calibration dataset into 4 segments.

The resulting SECV and standard error of prediction (SEP) were 0.30% and 0.29%, respectively, for the back-scattering measurement, while 0.20% and 0.23% for the transmission measurement. Seven factors were used in the PLS modeling for both measurements. The resulting concentration correlation plots for both measurements are shown in Fig. 7. Open and filled circles correspond to the calibration and prediction data, respectively. As shown in the figure, the calculated and predicted protein concentrations are closer to the ideal line in the transmission measurement. It clearly indicates that the transmission spectra of packed kernels contain more representative compositional information that eventually leads to a more accurate description for the variation of total protein concentration.

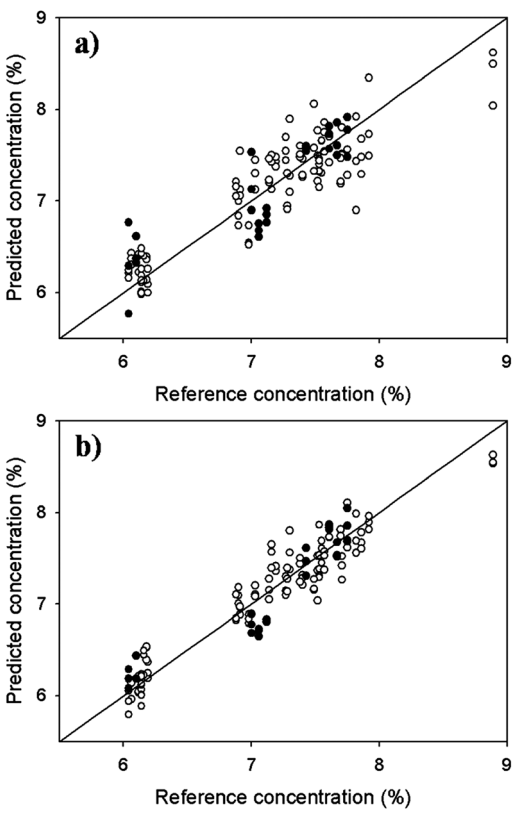

Fig. 7 Concentration correlation plots resulted from back-scattering (a) and transmission (b) measurements. Open and filled circles indicate the calibration and prediction data, respectively.

Additionally, to compare the reproducibility of back-scattering and transmission measurements, we calculated the standard deviation of three calculated or predicted protein concentrations for each sample, resulting in 45 standard deviations for 45 samples. The average standard deviations for back-scattering and transmission measurements were 0.18 and 0.11%, respectively. This result also indicates that triplicate spectra collected with random re-packing are more reproducible in the case of transmission measurement.

Conclusions

A highly potential transmission Raman measurement, performed directly through a packing of corn kernels to determine the protein concentration, has been demonstrated. Due to the ability of deeper sampling across the kernel packing, transmission Raman spectra were reproducible and provided a superior sample representation of the packed kernels. The demonstrated potential of transmission Raman spectroscopy could enhance its use for the analysis of diverse granular samples especially in the field of agriculture, feed and polymers, where application of this method is still in a relatively nascent stage.

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